

BCA SEMESTER V • INTRODUCTION TO MACHINE LEARNING • UNIT III

Evaluating an Estimator, Bayes' Estimator & Parametric Classification

Class 2: Statistical properties of estimators (Bias & Variance), Bayesian parameter estimation, and decision boundaries in parametric classification.

PRESCRIBED TEXTBOOK

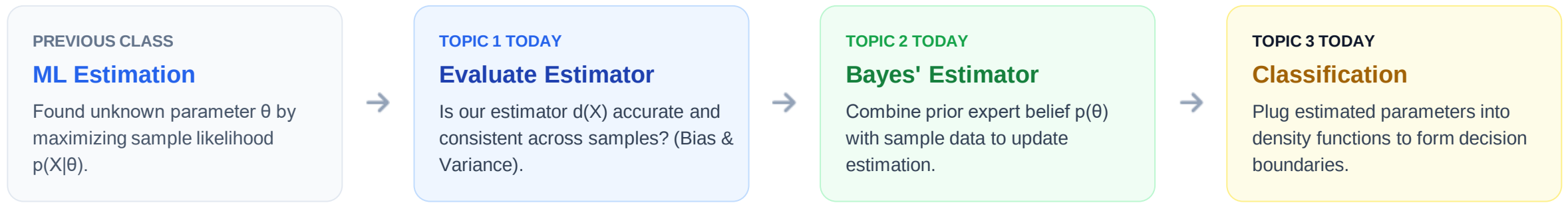
Ethem Alpaydin (MIT Press / PHI)

RELEVANT SECTIONS

Chapter 4: Sections 4.3, 4.4 & 4.5

Connecting Today's Lecture with Maximum Likelihood

From estimating unknown parameters to evaluating quality and making decisions.



💡 **Teacher Speaker Note:** "Last class, we used Maximum Likelihood to calculate a number for an unknown parameter θ . But how do we know if our formula is trustworthy? Today we evaluate estimators, look at Bayes estimation, and see how parameters classify real data." *Question for students:* "Can two different random samples from the same population give two different estimates?"

Why Do We Need to Evaluate an Estimator?

Estimators depend on random sample draws, leading to inherent variability.

The Estimation Problem

Let θ be an unknown true parameter of the underlying population distribution.

We draw a random sample $\mathbf{X} = \{x_1, x_2, \dots, x_N\}$ and compute a statistic $\mathbf{d} = \mathbf{d}(\mathbf{X})$ to estimate θ .

Because $\mathbf{X}$ is a random variable, our estimate $\mathbf{d}$ is also a **random variable** with its own distribution.

🎯 Question 1: Accuracy (Bias)

Does the average estimate over many random samples hit the true parameter value θ ?

↕ Question 2: Consistency (Variance)

How widely do estimates fluctuate from one drawn sample to another?

💡 **Teacher Speaker Note:** "Emphasize to students that $\mathbf{d}$ is a function of random data $\mathbf{X}$. If we change the sample, the estimate changes. Ask the class: 'If you survey 10 students today and 10 students tomorrow about average study hours, will the averages be identical?' This introduces sampling variation."

What is Bias? (Systematic Error)

Measuring whether an estimator hits the target on average.

$$\text{Bias}(d) = E[d] - \theta$$

Key Mathematical Symbols:

θ : The unknown true parameter of the population.

d : The estimator calculated from a given sample.

$E[d]$: The expected (average) value of the estimator across all possible samples.

An estimator is called Unbiased if: $\text{Bias}(d) = 0 \Rightarrow E[d] = \theta$



Bias measures the distance between the center of your estimates and the true parameter.

💡 **Teacher Speaker Note:** "Explain the word 'Expectation' $E[d]$ in simple terms: imagine taking 1,000 different samples and computing the average of those 1,000 estimates. If that average equals θ , your estimator has zero bias (unbiased)."

Understanding Bias with a Numerical Example

Suppose the true unknown parameter value is $\theta = 50$.

ESTIMATOR A

Low Bias Estimator

Estimates from 4 different samples:

48, 49, 51, 52

Average ($E[d]$): $(48+49+51+52) / 4 = 50.0$

Bias: $50.0 - 50 = 0$ (Unbiased)

✓ The average estimate is centered right on the true parameter $\theta = 50$.

ESTIMATOR B

High Bias Estimator

Estimates from 4 different samples:

40, 41, 39, 40

Average ($E[d]$): $(40+41+39+40) / 4 = 40.0$

Bias: $40.0 - 50 = -10.0$ (High Bias)

✗ The estimates are systematically shifted 10 units below the true value.

💡 **Teacher Speaker Note:** "Ask students: 'Notice that Estimator B numbers (39, 40, 41) are very close together. Does that mean Estimator B is good?' No! It is consistent, but it is consistently WRONG. That is the core idea of bias."

What is Variance? (Sampling Variability)

Measuring how much estimates fluctuate around their own expected value.

$$\text{Var}(d) = E[(d - E[d])^2]$$

Symbol Explanation:

d : Estimate from a single specific sample.

$E[d]$: The mean of all possible estimates.

$(d - E[d])^2$: Squared distance of an estimate from the center.

Interpretation: Variance measures sensitivity to the specific training dataset drawn.

LOW VARIANCE (Small spread around $E[d]$)



HIGH VARIANCE (Wide dispersion around $E[d]$)



💡 **Teacher Speaker Note:** "Remind students: variance has nothing to do with θ (the true value). It only measures how widely the estimates scatter around $E[d]$. A high variance estimator changes wildly every time you collect a new dataset."

Bias vs. Variance: Direct Comparison

Two independent sources of error in any statistical estimation procedure.

Property	Bias	Variance
Formula	$E[d] - \theta$	$E[(d - E[d])^2]$
What it measures	Distance to true target θ	Spread between samples
Error Type	Systematic error	Random sampling noise
Ideal Goal	Zero (Unbiased)	Minimum (Stable)



💡 **Teacher Speaker Note:** "Use the classic archery target analogy: the bullseye is θ . Bias is whether your shots center on the bullseye; variance is how close your arrows land next to each other."

Mean Square Error (MSE)

Combining Bias and Variance into a single metric for overall estimator quality.

ALPAYDIN'S EXPECTED LOSS (MSE)

$$r(d, \theta) = E[(d - \theta)^2] = \text{Var}(d) + (\text{Bias}(d))^2$$

1. Total Error $r(d, \theta)$

Expected squared difference between estimate d and true parameter θ .

2. Variance Component

Penalizes instability or excessive fluctuation across different random samples.

3. Squared Bias Component

Penalizes systematic shift away from the true population value.

💡 **Teacher Speaker Note:** "Notice that to minimize total error (MSE), we must keep BOTH variance and bias small. Sometimes an estimator with a tiny bias but huge variance is worse than one with small bias and small variance!"

Transition: What if We Have Prior Knowledge?

In Maximum Likelihood, θ is treated as an unknown constant fixed number.

⚠ The Limitation of Pure Sample Estimation

If our training sample size N is small, Maximum Likelihood can produce unreliable or extreme estimates.

However, domain experts or past experiments often provide **prior information** before observing the new dataset.

🧠 The Bayesian Solution

Treat the unknown parameter θ not as a fixed constant, but as a **random variable**!

Prior Knowledge $p(\theta)$ + Data Sample $X \rightarrow$ Updated Posterior $p(\theta|X)$

💡 **Teacher Speaker Note:** "Ask students: 'If you flip a coin 3 times and get 3 heads, does that mean the probability of heads is 100%?' MLE says yes, but your common sense (prior) says no. That common sense is what Bayesian estimation formalizes."

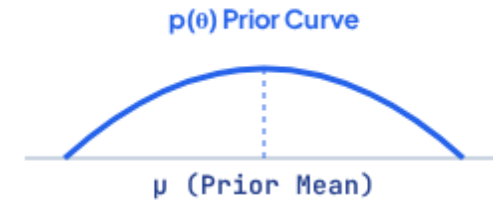
What is a Prior Distribution? $p(\theta)$

Formulating existing knowledge into a probability density over parameter values.

Definition of Prior Density $p(\theta)$

The **Prior Distribution** $p(\theta)$ expresses our degree of belief about the possible values of θ *before* observing the current sample data X .

- If we have strong prior knowledge, $p(\theta)$ has a narrow peak around expected values.
- If we have no prior preference (uninformative prior), $p(\theta)$ is flat over the domain.



The parameter θ is now treated as a random variable centered around prior mean μ .

💡 **Teacher Speaker Note:** "Make sure students understand the conceptual leap here: In Classical ML (previous class), θ was a single unknown fixed number. In Bayesian ML, θ is a random variable described by a probability distribution!"

Bayes' Rule for Parameter Estimation

Updating our belief about the parameter after seeing the sample data X .

$$p(\theta | X) = [p(X | \theta) \cdot p(\theta)] / p(X)$$

Posterior Probability: $p(\theta | X)$

Our updated belief about θ AFTER observing the sample data X .

Sample Likelihood: $p(X | \theta)$

Probability of drawing sample X given parameter θ (from previous class!).

Prior Probability: $p(\theta)$

Initial expert belief about θ BEFORE looking at sample data X .

Normalizing Constant: $p(X)$

Marginal probability of data X : $\int p(X|\theta)p(\theta) d\theta$.

💡 **Teacher Speaker Note:** "Write on the board: Posterior is proportional to Likelihood times Prior. State clearly that $p(X)$ does not depend on θ ; it is simply a constant that makes the area under the posterior integrate to 1."

What is the Bayes' Estimator?

Extracting a single optimal parameter value from the entire posterior distribution.

$$\theta_{\text{Bayes}} = E[\theta | X] = \int \theta p(\theta | X) d\theta$$

Plain English Meaning:

The **Bayes Estimator** is the mean (expected value) of the parameter under the posterior distribution.

Under the standard squared-error loss function, choosing the posterior mean minimizes the expected risk!

Why is this powerful?

Balances Prior & Evidence

Never relies solely on random sample flukes.

Handles Small Datasets

Prior stabilizes estimates when sample size N is tiny.

Converges to MLE as N grows

As data increases, sample evidence overwhelms the prior.

💡 **Teacher Speaker Note:** "Explain to students that the posterior distribution gives us a complete probability curve for θ . If we must output a single number for our ML model, the most natural choice is the center (mean) of that curve: $\theta_{\text{Bayes}} = E[\theta|X]$."

Comparison: MLE vs. MAP vs. Bayes Estimator

Three distinct paradigms for estimating parameters in machine learning.

Method	Mathematical Formula	Uses Prior?	Key Philosophy
Maximum Likelihood (MLE)	$\theta_{ML} = \operatorname{argmax}_{\theta} p(X \mid \theta)$	No	Finds parameter making observed data most likely.
Maximum A Posteriori (MAP)	$\theta_{MAP} = \operatorname{argmax}_{\theta} p(\theta \mid X)$	Yes	Finds peak (mode) of the posterior distribution.
Bayes' Estimator	$\theta_{Bayes} = E[\theta \mid X]$	Yes	Calculates expected value (mean) of the posterior.

💡 **Teacher Speaker Note:** "Point out the subtle difference between MAP and Bayes: MAP finds the highest point (mode) of the posterior curve, while Bayes finds the balance point (mean) of the posterior. For symmetric Gaussian distributions, MAP and Bayes give the exact same answer!"

Bayes Estimator Example: Normal-Normal Model

Estimating the mean of a Gaussian when the prior is also Gaussian.

Textbook Setup:

Observed samples drawn from Gaussian: $\mathbf{x}_t \sim \mathcal{N}(\theta, \sigma^2)$

Prior belief on θ is also Gaussian: $\theta \sim \mathcal{N}(\mu, \sigma^2)$

RESULTING BAYES ESTIMATE FORMULA:

$$\theta_{\text{Bayes}} = [N/\sigma^2 \cdot m + 1/\sigma^2 \cdot \mu] / [N/\sigma^2 + 1/\sigma^2]$$

where $m = \text{sample mean } \sum \mathbf{x}_t / N$

Intuition of the Formula:

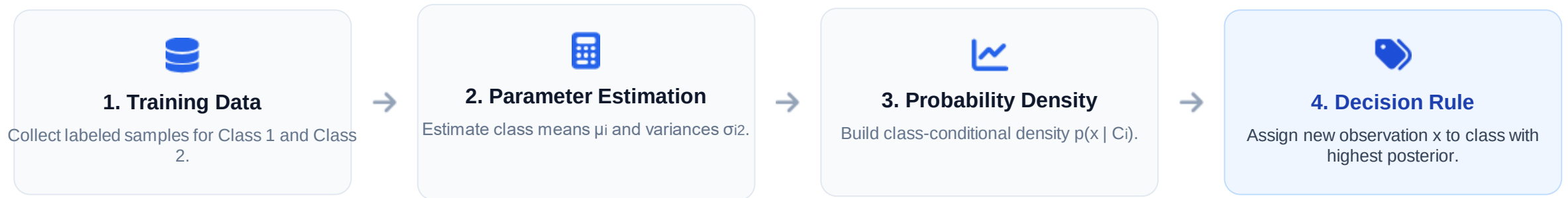
Weighted Average: θ_{Bayes} is a weighted blend between the sample mean m and the prior mean μ .

- If $N = 0$ (no data), $\theta_{\text{Bayes}} = \mu$ (pure prior).
- If $N \rightarrow \infty$ (huge sample), $\theta_{\text{Bayes}} \rightarrow m$ (pure MLE).

💡 **Teacher Speaker Note:** "Don't let students get frightened by the equation! Point out that each term is multiplied by its precision (1/variance). The more data you collect (large N), the more the formula ignores μ and trusts the sample mean m ."

Transition: From Estimation to Classification

Now that we know how to estimate parameters, how do we use them to classify?



💡 **Teacher Speaker Note:** "Connect this to Unit II (Bayesian Decision Theory): in Unit II, we assumed the probabilities $p(x|C_i)$ were handed down to us by magic. In Unit III, we actually CALCULATE them using sample parameters!"

What is Parametric Classification?

Assuming a known functional form for class distributions and fitting parameters.

The Core Assumption

In **Parametric Classification**, we assume each class C_i generates data according to a known parametric distribution (most commonly Gaussian).

Instead of learning arbitrary shapes, we only need to estimate a small fixed set of parameters:

Parameters $\Phi = \{ P(C_i), \mu_i, \sigma_i^2 \}$ for all classes i

Bayesian Classification Rule:

Choose C_i if: $P(C_i | x) > P(C_j | x)$

By Bayes' theorem, this is equivalent to:

$$p(x | C_i) \cdot P(C_i) > p(x | C_j) \cdot P(C_j)$$

💡 **Teacher Speaker Note:** "Highlight the huge advantage of parametric methods: we do not need to memorize millions of data points; we summarize the entire dataset of each class using just a mean and a variance!"

Gaussian Parametric Classification

Using the bell curve (Normal distribution) as our class-conditional model.

The Gaussian Model for Class C_i :

$$x | C_i \sim N(\mu_i, \sigma_i^2)$$

Estimated from training data:

μ_i : Sample mean of training samples belonging to Class i .

σ_i^2 : Sample variance of training samples in Class i .

$P(C_i)$: Proportion of samples belonging to Class i (Prior).

Discriminant Function $g_i(x)$:

Taking natural logs makes calculations simple:

$$g_i(x) = -\frac{1}{2} \ln(2\pi) - \ln(\sigma_i) - (x - \mu_i)^2 / (2\sigma_i^2) + \ln P(C_i)$$

We evaluate $g_i(x)$ for each class and choose the largest!

💡 **Teacher Speaker Note:** "Reassure students: taking the logarithm does not change the classification decision because log is a strictly increasing function. It turns difficult products and exponentials into simple additions!"

Decision Boundary: Equal Variances ($\sigma_1^2 = \sigma_2^2 = \sigma^2$)

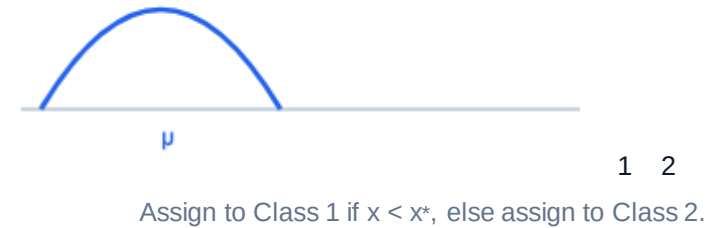
When both classes share the same spread, the decision boundary is simple and linear.

Single Threshold Boundary

If both classes have equal variance σ^2 and equal priors $P(C_1) = P(C_2)$, the quadratic squared terms cancel out!

$$x^* = (\mu_1 + \mu_2) / 2$$

The boundary is simply the **halfway midpoint** between the two class means.



Decision Boundary: Different Variances ($\sigma_{12} \neq \sigma_{22}$)

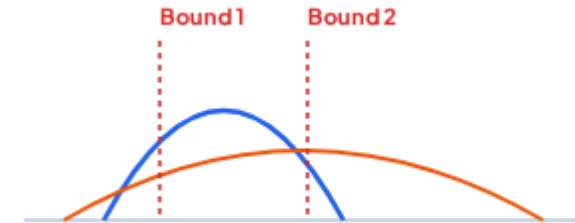
Unequal spreads produce quadratic equations, potentially yielding two decision thresholds.

Quadratic Decision Regions

When variances differ, the squared terms x^2 do not cancel out when equating $g_1(x) = g_2(x)$.

Yields a Quadratic Equation in $x \Rightarrow$ Can produce TWO boundaries!

The narrower distribution dominates near its mean, but the wider distribution overtakes it at extreme tails.



Equal variance = 1 boundary; Unequal variance = up to 2 boundaries.

💡 **Teacher Speaker Note:** "Draw students' attention to the diagram: because one distribution is tall and narrow while the other is short and wide, the curves intersect TWICE. This is a very common university exam question!"

Complete Lecture Summary: The Big Picture

Synthesizing the three pillars of today's class from Alpaydin Chapter 4.

PILLAR 1

Evaluating Estimators

- **Bias:** $E[d] - \theta$ (Systematic error).
- **Variance:** $E[(d - E[d])^2]$ (Sampling noise).
- **MSE:** $\text{Var}(d) + \text{Bias}^2$.

PILLAR 2

Bayesian Estimation

- Treats θ as random variable with prior $p(\theta)$.
- Updates belief via posterior $p(\theta|X)$.
- $\theta_{\text{Bayes}} = E[\theta|X]$ (Posterior mean).

PILLAR 3

Parametric Classification

- Assumes distribution shape (e.g. Gaussian).
- Equal variance $\rightarrow$ linear midpoint boundary.
- Unequal variance $\rightarrow$ quadratic (2 boundaries).

💡 **Teacher Speaker Note:** "Recap the conceptual progression: We learned how to measure estimator quality (Bias/Var), how to inject prior knowledge (Bayes), and how to apply those estimates to solve real classification tasks."

BCA Semester V • University Exam Questions

Master these questions from Chapter 4 of Ethem Alpaydin.

IMPORTANT SHORT QUESTIONS (2–3 MARKS)

1. Define Bias of an estimator.
2. Define Variance of an estimator.
3. State the relationship between MSE, Bias, and Variance.
4. What is a Prior Distribution $p(\theta)$?
5. Write the formula for Bayes' Estimator θ_{Bayes} .
6. What is the decision boundary when two Gaussian classes have equal variances?

DESCRIPTIVE QUESTIONS (5–8 MARKS)

1. Explain Bias and Variance of an estimator with suitable target diagrams.
2. Compare Maximum Likelihood (MLE), MAP, and Bayes Estimator with formulas.
3. Explain parametric classification using Gaussian models for equal vs. unequal variances.

5 Core Rules: (1) Bias is off-target; (2) Variance is scatter; (3) $\text{MSE} = \text{Var} + \text{Bias}^2$; (4) Bayes = Posterior mean; (5) Equal $\sigma = 1$ boundary, unequal $\sigma = 2$ boundaries.